

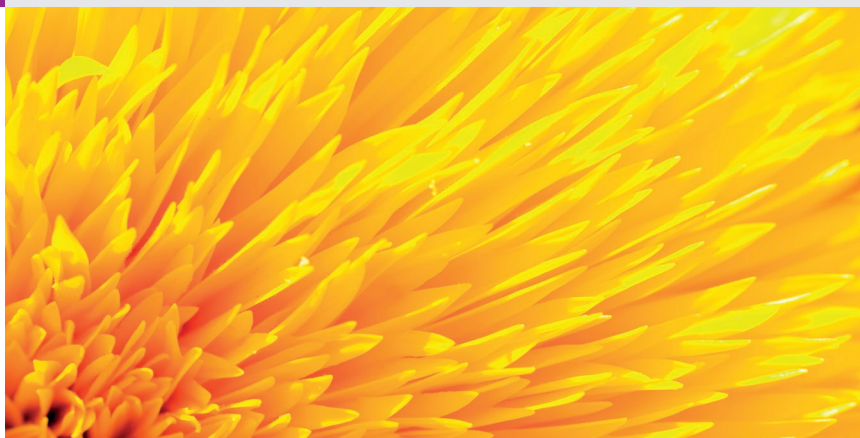
Acticire[®] MB

Active waxes



People make our name

Functionalized natural waxes



A 100% natural texturizing agent based on a complex of waxes functionalized in presence of polyglycerol, Acticire® MB forms a soft, smooth film that protects the skin from dehydration and ensures a comfortable skin feel throughout the day.

Natural waxes have interesting cosmetic properties: they form a barrier against water loss, they come from renewable sources and offer excellent safety profile.



Jojoba wax
Softness & easy spreading

+



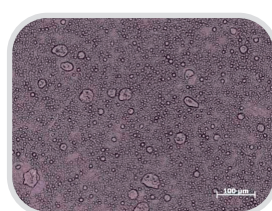
Mimosa wax
Film forming properties

+



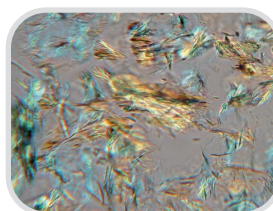
Sunflower wax
Excellent thermal stability

But they also have disadvantages. With the exception of jojoba wax, they demonstrate low sensoriality and poor compatibility with cosmetic emulsions.



At 5%
in O/W emulsion

Non-modified
sunflower/mimosa/
jojoba wax blend

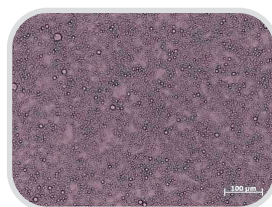


At 20% in apricot
kernel oil

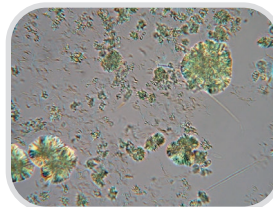


Rough skin feel

When applied on the skin



Acticire® MB



Smooth skin feel

Vegetable waxes protect the skin however, in their native form, they are difficult to formulate in emulsions. They also present a rough afterfeel on the skin.

In formulation, Acticire® MB forms small and rounder crystals that create a smooth film on the skin, leading to a softer touch.

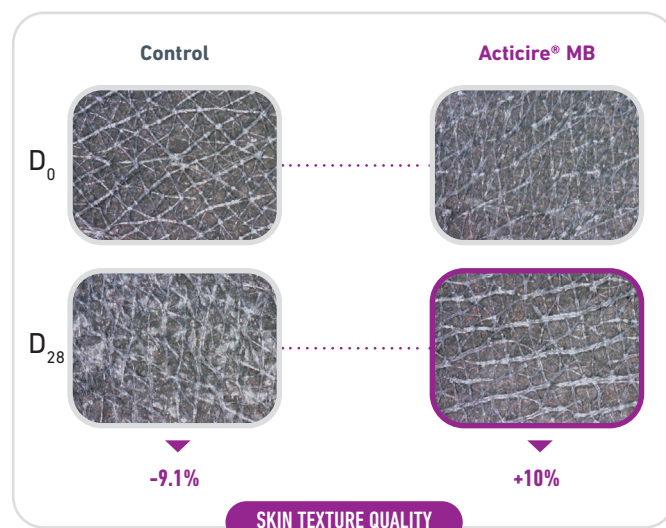
To obtain a functional and sensorial texturizing agent, Gattefossé has developed a patented technology of transesterification of solid and liquid waxes in presence of polyglycerol. Acticire® MB is perfectly stable in emulsion, and presents new sensorial advantages.

Naturally active textures

Moisturizing activity ⁽¹⁾

Several instrumental techniques have been used to evaluate Acticire® MB moisturizing properties in a simple O/W emulsion.

- **Trans Epidermal Water Loss (TEWL)** is directly correlated to the skin barrier function. A decrease of TEWL indicates a good protective effect of a formulation.
Acticire® MB offers a significant improvement of the water retention ability of the skin, compared to the non treated control area.
- **Corneometric measurement** is a genuine value of the skin's free water content.
The entire panel showed a very high moisturizing efficacy after treatment with Acticire® MB.
- **Skin microrelief** is a direct indicator of the level of the skin's moisturization. With age, or in dry skin, the microrelief becomes uneven and disorganized.
The quality of the skin surface of all volunteers is clearly improved after Acticire® MB application: visibly better organization and surface unaltered by scales.



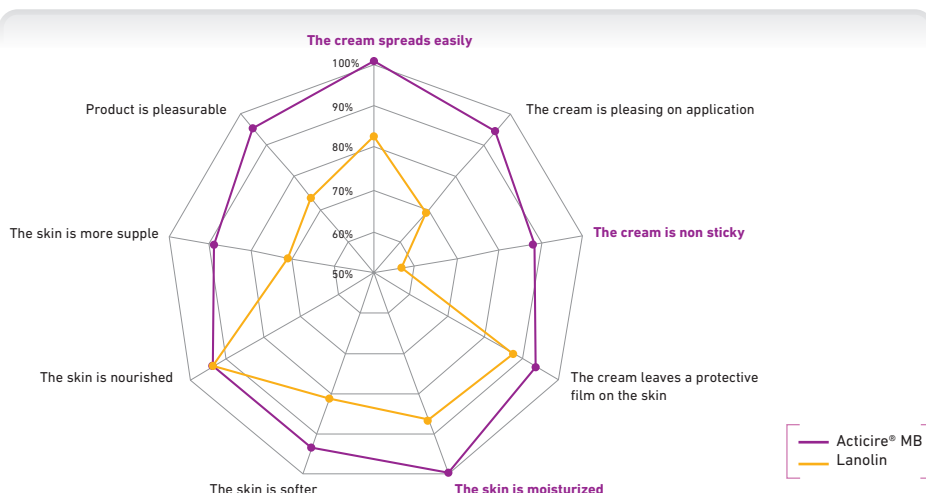
Extremely comfortable sensation ⁽²⁾

Natural formulations, containing a majority of vegetable oils, frequently present a rich feel during application yet a dry afterfeel within several minutes, giving an impression of lack of moisturization and comfort. This is principally due to the absence of protecting film on the surface of the skin.

Acticire® MB improves application and afterfeel by forming a perceptible yet **light film on the skin**, thus providing a **long lasting comfort sensation**. Capable of absorbing up to 6 times its own weight in water, Acticire® MB forms a soufflé-like structure that presents an elastic, cushion soft feel.

Acticire® MB also reduces the sensory impact of layering on the tacky afterfeel when used at 2% in a cream.

Acticire® MB was evaluated by a panel of users over a period of 7 days.



Acticire® MB:

- Very good overall satisfaction level
- Strong moisturizing effect confirmed by 100% of the panelists

Acticire® MB provides softness and spreadability upon application, with a comfortable skin feel that lasts throughout the day.

⁽¹⁾ Clinical study, 4 weeks, 25 volunteers with dry skin, aged 28 to 68. Twice daily application of an emulsion containing 5% of Acticire® MB.

⁽²⁾ Two formulations, one containing 5% deodorized lanolin and the other 5% Acticire® MB were tested on a panel of 16 women, over a period of 7 days. Each cream is used twice daily, morning and evening.

A breakthrough in cosmetic applications

Acticire® MB can be formulated in all kinds of emulsion and anhydrous formulation, except transparent systems. It is generally added to the oil phase where it is solubilized by melting at 70°C.

Emulsions

Acticire® MB can be used in W/O and O/W emulsions as a **texture agent**.

Acticire® MB plays an important role in **improving the compatibility with different types of oils**, especially in W/O emulsions. It is particularly compatible with natural, PEG-free emulsifiers, including sugar esters, polyglyceryl esters and anionic emulsifiers. Acticire® MB also helps to **reduce the tackiness and greasiness** brought by some lipophilic components.

Anhydrous textures

Acticire® MB is particularly interesting for its capacity to **disperse a water phase into anhydrous formulations**, due to the presence of amphiphilic molecules. It allows to introduce hydrophilic components, such as actives in various textures (massage balms or lipsticks) and therefore improves efficacy and creates new sensations.

► Balms

Used at 5% in a balm formula, Acticire® MB allows to **incorporate up to 15% of an aqueous phase**. It forms an original texture, similar to butter, but with new sensorial properties.

Unlike lanolin, which also has water absorption capacity, Acticire® MB allows to obtain a very **stable formulation** at high temperatures.

Anhydrous-like balm	Heat stability 40°C 3 months
5% lanolin 15% water	Unstable
5% Acticire® MB 15% water	Stable

► Lipsticks

Used at 5% in a lipstick base, Acticire® MB is able to **disperse homogeneously** 2% of a water phase (colored with hydrophilic dye in our example for an optimal visualization), giving an **even color without visible droplets**.

In addition, it also improves stability, application properties and texture.

Lipstick with 2% water phase
(blue aqueous food colorant)



Suncare applications

Acticire® MB is a true ally for suncare:

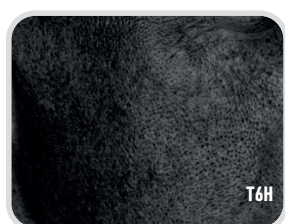
- **Improves the sensoriality of suncare formulas:**

Reduces unpleasant sensory attributes such as greasiness and limits the sensory gap between temperate and tropical conditions.

- **Improves the durability of the film:**

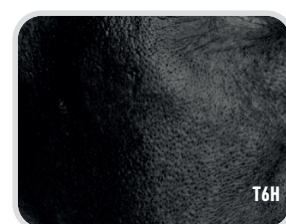
Qualitative and quantitative tests performed with a UV camera.

Without
Acticire® MB



The film is lighter than the other one, freckles start to appear

With
Acticire® MB



The film is darker: UVA protection is greater



Acticire® MB

is a texturizing and moisturizing ingredient, designed to upgrade any cosmetic formulations. 100% natural, based on a functionalized complex of jojoba, mimosa and sunflower waxes, Acticire® MB brings durable moisturization and comfort to the skin.

- Combines the sensorial and functional advantages of liquid emollients, butters and waxes
- Stabilizes and improves the sensoriality of natural formulations including sun care
- Allows the creation of original textures

Claims

- Active texturizing agent
- Brings lasting moisturization and comfort
- Enhances sensorial features
- Improves formulations stability
- Enables innovative textures

Applications

- Skin care: O/W and W/O emulsions, lip and massage balms, natural moisturizing lines
- Sun care: formulations with high UV filter levels
- Make-up: foundations, powders, innovative lipsticks and balms

Sensoriality

- Softness and spread upon application
- Comfortable skin feel

Use level

1-5%

INCI name: Jojoba Esters (and) Helianthus Annuus (Sunflower) Seed Wax (and) Acacia Decurrens Flower Wax (and) Polyglycerin-3

Worldwide approval for cosmetic applications
Listed on IECIC (Inventory of Existing Cosmetic Ingredients in China) 2015

COSMOS approved

As a user of palm oil and palm kernel oil derivatives, Gattefossé is sensitive to the environmental impact of palm tree growing and participates in the development of sustainable cultivation, providing its customers with environmentally responsible ingredients.

Gattefossé is a proactive member of RSPO (Roundtable on Sustainable Palm Oil) since 2010. In 2013, Gattefossé achieved the RSPO supply chain certification. Since then, Gattefossé offers products certified RSPO Segregated, and more recently, products certified RSPO Mass Balance. Acticire® MB is certified Mass Balance.

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